

ASSIGNMENT - 3

1. Determine the multiplicative inverse modulo 26 for each of the following integers.

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

3⁻¹

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1)$$

$$1 = 3 + [26 + 3(-8)](-1)$$

$$2 + 26(-1) + 3(8) = 1$$

$$3(9) + 26(-1) = 1$$

$$3^{-1} = \underline{9}$$

5⁻¹

$$26 = 5(5) + 1$$

$$26 + 5(-5) = 1$$

$$26 - 5 = 21$$

$$5^{-1} = \underline{21}$$

$$1 = (1) [26 + 5(-5)] + (5)(1) + 26$$

7⁻¹

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$26 + 7(-3) + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + 5(2) = 1$$

$$26 + 7(-5) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$7^{-1} = \underline{\underline{5}}$$

$$9^{-1}$$

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + [26(-1) + 9(2)] = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = \underline{\underline{3}}$$

$$11^{-1}$$

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$(8-1)26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$(1-8)26 + 11(-2) + 11(-1) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-2) + 11(-1) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$11^{-1} = \underline{\underline{19}}$$

$$15^{-1}$$

$$(1) \text{PI } 26 = 15(1) + 11$$

$$(2) \text{F } 15 = 11(1) + 4$$

$$(1) \text{S } 11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + [11 + 4(-2)](-1) = 11(5) + 15$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + [26 + 15(-1)](-2) + [15 + 11(-1)](2) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + [26 + 15(-1)](-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$15^{-1} = 7$$

$$17^{-1}$$

$$26 = 17(1) + 9$$

$$17 = 9(1) + 8$$

$$9 = 8(1) + (17 - 9)$$

$$9 + 8(-1) = 1$$

$$26 + 17(-1) + [17 + 9(-1)](-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-2) + (26 + 17(-1))(1) = 1$$

$$26 + 17(-2) + 26(1) + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = 23$$

$$17^{-1} = 23 //$$

19-1

$$26 = 19(1) + 7$$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$19 + 7(-2) + [7 + 5(-1)](-2) = 1$$

$$19 + 7(-2) + 7(-2) + 5(2) = 1$$

$$19 + [26 + 19(-1)](-4) + [7 + 7(-2)]2 = 1$$

$$19 + 26(-4) + 19(4) + 19(2) + 7(-4) = 1$$

$$19 + 26(-4) + 19(4) + 19(2) + [26 + 19(-1)](-4) = 1$$

$$19 + 26(-8) + 19(10) = 1$$

$$19(\underline{11}) + 26(-8) = 1$$

$$19^{-1} = \underline{11}$$

21-1

$$26 = 21(1) + 5$$

$$5 = 26 + 21(-1)$$

$$(1)21 + 21 = 0 \quad 5(4) + 1$$

$$(1)21 + 5(-4) = 1$$

$$21 + [26 + 21(-1)](-4) = 11 \cdot 8 = 8$$

$$21 + 26(-4) + 21(4) = 11 \cdot 8 + 8$$

$$21(\underline{5}) + 26(-4) = 1 + 8 \quad + (1)21 + 26$$

$$21^{-1} = \underline{5} \quad (1)21 + 26 + (1)21 + 26$$

$$1 = (1)21 + (1)26 + (1)21 + 26$$

$$1 = (8)21 + (5)26$$

$$21 \cdot 8 = 168$$

$$26 \cdot 5 = 130$$

$$23^{-1}$$

$$26 = 23(1) + 3$$

$$3 = 26 + 23(-1)$$

$$23 = 3(7) + 2$$

$$2 = 23 + 3(-7)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$26 + 23(-1) + [23 + 3(-7)](-1) = 1$$

$$26 + 23(-1) + 23(-1) + 3(7) = 1$$

$$26 + 23(-2) + [26 + 23(-1)](7) = 1$$

$$26 + 23(-2) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$23^{-1} = \underline{\underline{17}}$$

$$25^{-1}$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$25^{-1} = \underline{\underline{25}}$$

2. choose any string of your choice to encrypt using the affine cipher with key values:

Multiplier 5 and Additive 8.

Plaintext: SHRUTHA

Formula: $C = (P \cdot \text{Multi} + \text{Add}) \bmod 26$

$$S = (18 \times 5 + 8) \bmod 26 = 98 \bmod 26 = 20 = U$$

$$H = (7 \times 5 + 8) \bmod 26 = 43 \bmod 26 = 17 = R$$

$$R = (17 \times 5 + 8) \bmod 26 = 93 \bmod 26 = 15 = P$$

$$U = (20 \times 5 + 8) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$T = (19 \times 5 + 8) \bmod 26 = 103 \bmod 26 = 25 = Z$$

$$H = (7 \times 5 + 8) \bmod 26 = 43 \bmod 26 = 17 = R$$

$$A = (0 \times 5 + 8) \bmod 26 = 8 \bmod 26 = 8 = I$$

URPEZRI

3. encrypt 'HI' using affine cipher with $A = 11$, $B = 5$

$$C = (P \cdot a + b) \bmod 26$$

$$C = (HI \cdot 11 + 5) \bmod 26$$

$$H = (7 \times 11 + 5) \bmod 26 = 82 \bmod 26 = 4 = E$$

$$I = (8 \times 11 + 5) \bmod 26 = 93 \bmod 26 = 15 = P$$

EP

4. The ciphertext UCR was encrypted using the affine function $9x + 2 \pmod{26}$. Find the plaintext

$$C = (9P + 2) \pmod{26}$$

$$P = a^{-1} (C - b) \pmod{26}$$

$$9^{-1} =$$

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$8 = 26 + 9(-2)$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$9^{-1} = 3$$

$$U = 3(20 - 2) \pmod{26} = 54 \pmod{26} = 28 \pmod{26} = 2 = C$$

$$C = 3(2 - 2) \pmod{26} = 0 \pmod{26} = 0 = A$$

$$R = 3(17 - 2) \pmod{26} = 45 \pmod{26} = 19 = T$$

CAT

5. Decrypt the ciphertext using affine cipher. KDYXJWDEXZY using the key 3 (mult) & 4.

$$P = a^{-1} (C - b) \pmod{26}$$

$$3^{-1} =$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$3 + [26 + 3(-8)](-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$3(9) + 26(-1) = 1$$

$$3^{-1} = 9$$

$$K = 9(10-4) \bmod 26 = 2 \bmod 26 = 2 = C$$

$$D = 9(3-4) \bmod 26 = 17 \bmod 26 = 17 = R$$

$$Y = 9(24-4) \bmod 26 = 24 \bmod 26 = 24 = Y$$

$$X = 9(23-4) \bmod 26 = 15 \bmod 26 = 15 = P$$

$$J = 9(9-4) \bmod 26 = 19 \bmod 26 = 19 = T$$

$$U = 9(20-4) \bmod 26 = 14 \bmod 26 = 14 = O$$

$$W = 9(22-4) \bmod 26 = 6 \bmod 26 = 6 = G$$

$$D = 9(3-4) \bmod 26 = 17 \bmod 26 = 17 = R$$

$$E = 9(4-4) \bmod 26 = 0 \bmod 26 = 0 = A$$

$$X = 9(23-4) \bmod 26 = 15 \bmod 26 = 15 = P$$

$$Z = 9(25-4) \bmod 26 = 7 \bmod 26 = 7 = H$$

$$Y = 9(24-4) \bmod 26 = 24 \bmod 26 = 24 = Y$$

CRYPTOGRAPHY

6. Use $K_1 = 13$ (Multi), $K_2 = 4$ (then though it is not a valid key - key to encrypt). Encrypt the string 'input' and 'alter' using above key & find what is the issue.

$$C = (P \cdot \text{Multi} + \text{Add}) \bmod 26$$

$$C = (P \cdot a + b) \bmod 26$$

$$a = 13, \quad b = 4$$

13 is not coprime of 26

so we can't find inverse of 13 mod 26.

therefore it is not a valid key for affine cipher

For 'input'

$$I = (8 \times 13 + 4) \bmod 26 = 108 \bmod 26 = 4 = E$$

$$N = (13 \times 13 + 4) \bmod 26 = 173 \bmod 26 = 17 = R$$

$$P = (15 \times 13 + 4) \bmod 26 = 199 \bmod 26 = 17 = R$$

$$U = (20 \times 13 + 4) \bmod 26 = 264 \bmod 26 = 4 = E$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

ERROR

For 'alter'

$$A = (0 \times 13 + 4) \bmod 26 = 4 \bmod 26 = 4 = E$$

$$L = (11 \times 13 + 4) \bmod 26 = 147 \bmod 26 = 17 = R$$

$$T = (19 \times 13 + 4) \bmod 26 = 251 \bmod 26 = 17 = R$$

$$E = (4 \times 13 + 4) \bmod 26 = 56 \bmod 26 = 4 = E$$

$$R = (17 \times 13 + 4) \bmod 26 = 225 \bmod 26 = 17 = R$$

ERROR

Both the plaintext will produce the same ciphertext. The encryption is not one to one and decryption is not possible as 13 is not a valid key.

7. Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$$

$$\det(K) = 3 \times 7 - 2 \times 5$$

$$= 21 - 10 = 11$$

$$11^{-1} = 26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$3 = 11 + 4(-2)$$

$$4 = 26 + 11(-2)$$

$$[26 + 11(-2)] + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + (26 + 11(-2))2 = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

Adjoint of the Matrix

$$\begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 133 & -38 \\ 95 & 57 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8. Find the inverse of the following matrix whose entries are considered module 26:

$$K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \text{ mod } 26$$

$$\det K = (1)(3) - (2) \times 13 = 7$$

$$7^{-1}$$

$$26 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(52) + 5(2) = 1$$

$$26 + 7(-5) + 5(2) = 1$$

$$26 + 7(-5) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

Adjoint is

$$\text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

$$X = 01 = 25 \text{ hour 25}$$

$$Y = 05 = 25 \text{ hour 05}$$

9. Is the key valid?

$$\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$$

For a key to be valid - its determinant should be coprime and shouldn't be equal to 0

$$\det(K) = 2(21) - 14(13)$$

$$= 42 - 182$$

$$= -140$$

$$-140 \bmod 26 = 16$$

$$\det K = 16 \bmod 26$$

16 is not coprime to 26.

Therefore it will not have an inverse mod 26

$\therefore$ The Key $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$ is not valid.

10. Encrypt using 2×2 Hill cipher for the plaintext 'SOLVED' using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$3 \times 1 - 7 \times 8 = -23 = 3 \text{ which is coprime}$$

SOLVED

$$C = K \times P$$

$$P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 280 \end{bmatrix} \bmod 26$$

$$166 \bmod 26 = 10 = K$$

$$280 \bmod 26 = 20 = V$$

$$\begin{matrix} L \\ V \end{matrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \pmod{26}$$

$$201 \pmod{26} = 19 = T$$

$$308 \pmod{26} = 22 = W$$

$$\begin{matrix} E \\ D \end{matrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \pmod{26}$$

$$36 \pmod{26} = 10 = K$$

$$61 \pmod{26} = 9 = J$$

SOLVED $\rightarrow$ KUTWKJ